



Sima Barzegar

Date of birth: 23/03/1988 | **Place of birth:** | **Nationality:** Iranian | **Gender:** Female |

Email address: sima.barzegar@bsc.es | **Website:** <https://www.bsc.es/barzegar-sima> | **LinkedIn:** <https://www.linkedin.com/in/sima-barzegar/> |

Address: C. Jordi Girona 1-3, 08034, Barcelona, Spain (Work)

EDUCATION AND TRAINING

22/09/2006 – 21/06/2010 Urmia, Iran

BACHELOR Islamic Azad University Branch of Urmia

Electrical engineering _ electronics

Address Ayatollah Rafsanjani Complex, Islamic Azad University, Urmia, West Azerbaijan, Urmia, Iran |

Website <https://urmia.iau.ir/en> | **Field of study** Electricity and energy , Electronics and automation | **Final grade** 16.15 out of 20

30/11/2009 – 30/11/2011 Tabriz, Iran

MASTER OF SCIENCE Sahand University of Technology

In this thesis a new structure for Micro-Electro-Mechanical variable capacitor by varying common areas of plates, designed and simulated. The structure is designed using thermal-electrical activators to have a variable and tunable capacitance. The Latch structure is used to discretize structure movement and hold movable plate in steps with interval of one micrometer. The capacitor manufactured using polysilicon due to its good properties for these structures and its wide application in MEMS fabrication. ANSYS and HFSS FEM softwares are used to study the high frequency operation of structure and also mechanical operation and hence optimizing the structure. Operational parameters of structure are extracted using these FEM simulations.

Website <http://www.sut.ac.ir/> | **Field of study** Electronics and automation , Electricity and energy | **Final grade** 16.81 out of 20 |

Thesis Design and simulation of tunable capacitor based on RF MEMS technology

Link <https://doi.org/10.1007/s11277-015-2514-z>

14/03/2019 – 02/06/2022 Barcelona, Spain

PHD Universitat Politècnica de Catalunya

The first objective of my thesis focuses on the reliability of the optical network and proposes methods for health analysis related to Quality of Transmission degradation. Such degradation is produced by soft-failures in optical devices and fibers in core and metro segments of the operators' transport networks. Here, we compare estimated and measured QoT in the optical transponder by using a QoT tool based on GNPY. We show that the changes in the values of input parameters of the QoT model representing optical devices can explain the deviations and degradation in performance of such devices. We use reverse engineering to estimate the value of those parameters that explain the observed QoT. We show by simulation a large anticipation in soft-failure detection, localization and identification of degradation before affecting the network. Finally, for validating our approach, we experimentally observe the high accuracy in the estimation of the modeling parameters.

The second objective focuses on multilayer optical networks, where lightpaths are used to connect packet nodes thus creating virtual links. We study how lightpaths can be managed to provide enough capacity to the packet layer without detrimental effects in their Quality of Service. Such management must be as autonomous as possible to minimize human intervention. We study the autonomous operation of optical connections based on digital subcarrier multiplexing (DSCM). We propose the combination of two modules running in the optical node and in the optical transponder activate and deactivate subcarriers to adapt the capacity of the optical connection to the upper layer packet traffic. The module running in the optical node is part of our Intent-based Networking (IBN) solution and implements prediction to anticipate traffic changes. Our study demonstrates the feasibility of DSCM autonomous operation and shows large cost savings in terms of energy consumption and provides a guideline to help vendors and operators to adopt the proposed solutions.

The final objective targets at automating packet layer connections. Automating the capacity required by PKCs can bring further cost reduction to network operators, as it can limit the resources used at the optical layer. However, such

automation requires careful design to avoid any QoS degradation, which would impact Service Level Agreement in the case that the packet flow is related to some customer connection. We study autonomous packet flow capacity management. We apply RL techniques and propose a management lifecycle consisting of three different phases: 1) a self-tuned threshold-based approach for setting up the connection until enough data is collected; 2) RL operation based on models pre-trained with generic traffic profiles; and 3) RL operation based on models trained with the observed traffic. The proposed lifecycle provides remarkable performance from the starting of the connection and it shows the robustness while facing changes in traffic.

Address Plaça Eusebi Güell, 6, 08034, Barcelona, Spain | **Website** <https://www.upc.edu/ca> |

Field of study Information and Communication Technologies (ICTs) not further defined , Software and applications development and analysis , Inter-disciplinary programmes and qualifications involving Information and Communication Technologies (ICTs) |

Final grade Cum laude | **Thesis** Autonomous and reliable operation of multilayer optical networks

Link <https://orcid.org/0000-0003-1916-7217>

● **WORK EXPERIENCE**

UNIVERSITY RESEARCH ASSISTANT – DEPARTMENT OF ELECTRONICS ENGINEERING, FACULTY OF ENGINEERING, ISLAMIC AZAD UNIVERSITY URMIA BRANCH – 30/11/2007 – 30/11/2008 – URMIA, IRAN

Calibration of biomedical sensors, Robotics, electronics circuits with AVR, ARM and etc.

ELECTRONICS ENGINEER – SAYTAK – 31/12/2008 – 31/08/2011 – URMIA, IRAN

Designer of electronic circuits, AVR programmer, ARM programmer

ELECTRONICS ENGINEER – TELECOM COMPANY OF URMIA – 31/05/2009 – 31/01/2010 – URMIA, IRAN

Apprenticeship, assigned project based on controlling and monitoring the usage of customers.

UNIVERSITY RESEARCH ASSISTANT – SAHAND UNIVERSITY OF TECHNOLOGY – 30/11/2009 – 30/11/2012 – TABRIZ, IRAN

Researcher at department of electrical and electronics engineering. I have been working on designing and simulating the radio frequency (RF) sensors and RF capacitors as well as RF MEMS thermal-electrical activators and RF MEMS Varactors.

UNIVERSITY RESEARCH ASSISTANT – UNIVERSITAT POLITÈCNICA DE CATALUNYA – 14/03/2019 – 02/06/2022 – BARCELONA, SPAIN

Business or Sector: Information and communication

Applying machine learning, reinforcement learning in beyond 5G networks, as well as data collection for machine learning environments. python programming languages and simulation environment using especial tools. I have been involved in different European projects like **H2020 B5G-OPEN**(5G-PPP), **H2020 METRO-HAUL**(5G-PPP) and I have been working in these projects in 5G technologies (including fixed and mobile networks) and end-to-end and cloud computing, particularly with regard to the use of monitoring data and its processing through machine learning techniques and developing digital twins by using machine learning techniques. I have published several jurnal and conference papers and presented them. I have written technical and scientific reports. I have leaded master students in the research projects and have collaborated with national and international groups (industry and university). I have worked in a multidisciplinary group during my PhD study.

UNIVERSITY RESEARCH ASSISTANT – UNIVERSITAT POLITÈCNICA DE CATALUNYA – 03/06/2022 – 31/01/2025 – BARCELONA, SPAIN

Business or Sector: Professional, scientific and technical activities | **Department:** Computer Architecture |

Email: sima.barzegar@upc.edu | **Website:** <https://www.upc.edu/ca>

My research has focused on the autonomous and reliable operation of multilayer optical networks, where I have developed innovative solutions to enhance the quality of transmission, optimize network resource allocation, and automate operations. My doctoral work was recognized with the Special Doctoral Award in 2024.

During my academic career, I have successfully managed and collaborated on several **H2020** and **HORIZON European research projects**, such as **METRO-HAUL**, **B5G-OPEN**, **SEASON**, and **DESIRE-6G**, where I have led activities related to network and service automation. These projects involved managing project deliverables, coordinating with international partners, and ensuring timely and successful execution of project objectives. I have worked closely with a diverse set of partners, including telecom operators (**Telefónica**, **British Telecom**, and **Telecom Italia**), technology vendors (**Nokia**, **Ericsson**, **Infinera**, **Cisco**), and research institutions across Europe.

In addition to my research expertise, I have organized multiple workshops to promote the role of women in telecommunications and have served as a reviewer for international journals. My experience coordinating complex multi-institutional collaborations aligns closely with the requirements of this role. I am fluent in English and have a working knowledge of Spanish, which will enable me to communicate effectively with both local and international team members.

INTERNATIONAL HPC TRAINING PROGRAM OFFICER – BARCELONA SUPERCOMPUTING CENTER – 01/02/2025 –
Current – BARCELONA, SPAIN

Business or Sector: Education | **Department:** Directors | **Email:** sima.barzegar@bsc.es | **Website:** <https://www.bsc.es/es>

- Lead the Training Coordination and Collaboration work package for the CASTIEL 3 project, focusing on roadmap harmonisation and ecosystem-wide coordination across European HPC stakeholders.
- Previously led the Training, Twinning, and Mentoring work package for the CASTIEL 2 project, delivering training strategy and collaboration initiatives across European HPC centres.
- Design and implement specialised training programmes for scientists across Europe, collaborating with international HPC centres, industry, and academic partners.
- Coordinate cross-functional team activities within European-funded projects, ensuring alignment with strategic objectives and timely delivery of milestones.
- Drive international collaboration initiatives, facilitating coordination and knowledge exchange across European HPC ecosystem.
- Co-author and contribute to successful Horizon Europe and EuroHPC project proposals, supporting the acquisition of competitive research funding.
- Oversee the development and quality assurance of training materials, identifying content gaps and ensuring technical consistency with project objectives.
- Organise and coordinate training events, workshops, and seminars for diverse audiences, from early-career researchers to advanced users.
- Prepare and deliver project documentation, reports, presentations, and official deliverables in line with EU project requirements.
- Implement structured project tracking and monitoring processes to improve visibility and delivery of key outputs.
- Lead synchronisation efforts with international partners to harmonise training curricula across institutions.

● **SKILLS**

microsoft office | python | c++ | Deeplearning | machine learning | linux | windows | reinforcement learning | Data analysis | data science | artificial intelligent | Team-work oriented | Research and analytical skills | Creativity | Motivated | Critical thinking | Organizational and planning skills | Good listener and communicator | Analytical skills | Responsibility | Flexibility | Decision-making | Presenting | Ability to Work Under Pressure | Detail-Oriented | Research | perform project management | disseminate results to the scientific community | Problem-solving | Microsoft Word | Google Drive | Ability to multitask, plan and prioritise workload to ensure deadlines are met | strategic planning | think proactively | Responsible, highly educated, and passionate about work. | project management

● **LANGUAGE SKILLS**

Mother tongue(s): **AZERBAIJANI** | **PERSIAN**
Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
TURKISH	C1	C1	B2	B2	B2
SPANISH	C1	C1	B2	B2	B2
ENGLISH	C1	C1	C1	C1	C1

● CONFERENCES AND SEMINARS

30/11/2017 – 30/11/2018 International Conference on Transparent Optical Networks (ICTON)

Adaptation of the residual signal for filter failure detection in scenarios with multiple filter types

To monitor the optical spectra of outgoing links, Optical Spectrum Analyzers (OSA) can be deployed in the optical nodes acquiring the optical spectra of outgoing links and analyzing optical signals to detect those soft-failures that affect the shape of the spectrum, like Filter Shift and Filter Tightening. In this paper, we rely on the residual-based approach spectra analysis that uses a set of residual signals calculated by subtracting the measured signal by the OSA from an expected version of the signal. In this paper, we assume a network scenario, where nodes with optical filters of different characteristics (e.g., 2nd, 3rd, and 4th order Gaussian) are installed. In this scenario, the possibility of generating complete data set for machine learning training purposes that include every combination of filter and failure is difficult to achieve. Instead, we propose an adaptation mechanism that transform the expected effects of every filter type thus, emulating single filter-type networks. Exhaustive experiments are carried out to show the performance of the proposed adaptation under filter failures. © 2019 IEEE.

Link <http://www.scopus.com/inward/record.url?eid=2-s2.0-85073073130&partnerID=MN8TOARS>

30/11/2018 International Conference on Transparent Optical Networks

Soft-failure localization and time-dependent degradation detection for network diagnosis

In optical networks, degradation of the Quality of Transmission (QoT) can be the outcome of soft-failures in optical devices, like Optical Transponders, Wavelength Selective Switches (WSS) and Optical Amplifiers (OA). In this paper, we assume time-dependent degradations on ROADMs and OAs. Specifically, several degradations are considered: i) the noise figure can increase linearly over time due to the aging of the components; ii) the maximum of optical output power of the amplifiers can decrease because of the degradation in the pump lasers of the EDFAs; iii) aging effects, e.g., due to fiber splices; and iv) the OSNR can vary caused by frequency drift of WSSs due to temperature variations. Our proposal for degradation detection and soft-failure localization includes algorithms that are able to detect and localize the degradation in early stages and facilitate network diagnosis. In addition, we propose an architecture where the control plane consists of a network controller, a Monitoring and Data Analytics system and a QoT tool based on GNPys that are interconnected with each other.

Link <http://www.scopus.com/inward/record.url?eid=2-s2.0-85092493924&partnerID=MN8TOARS>

30/11/2018 OPTICAL FIBER COMMUNICATIONS CONFERENCE AND EXPOSITION (OFC)

Soft-failure localization and device working parameters estimation in disaggregated scenarios

A soft-failure localization and key working parameters estimation system is proposed for network diagnosis and maintenance. We show that a double analysis of monitoring data and estimated working parameters greatly anticipates degradations. OFC 2020 © OSA 2020 © 2020 The Author(s)

Link <http://www.scopus.com/inward/record.url?eid=2-s2.0-85089943557&partnerID=MN8TOARS>

30/11/2020 Optical Fiber Communications Conference and Exhibition, San Francisco, CA, USA

Reliable and Accurate Autonomous Flow Operation based on Off-line Trained Reinforcement Learning

A RL agent trained offline for reliability and able to refine its policies during online operation is proposed. Results for three illustrative flow automation use cases show remarkable performance with extraordinary adaptability to changes. © 2021 OSA.

Link <https://ieeexplore.ieee.org/document/9489915>

30/11/2020 Optical Fiber Communications Conference and Exhibition, San Francisco, CA, USA

Combining Long-Short Term Memory and Reinforcement Learning for Improved Autonomous Network Operation

A combined LSTM and RL approach is proposed for dynamic connection capacity allocation. The LSTM predictor anticipates periodical long-term sharp traffic changes and extends short-term RL knowledge. Numerical results show remarkable performance. © 2021 OSA.

Link <http://www.scopus.com/inward/record.url?eid=2-s2.0-85112083990&partnerID=MN8TOARS>

Reinforcement Learning -based Autonomous Multilayer Network Operation

We propose an Intent-Based Networking solution based on Reinforcement Learning (RL). RL-based intents with specific operational objectives and coordinated operation between network layers are developed. Results show service assurance with automatic adaptation to changing traffic.

Link <https://ieeexplore.ieee.org/document/9333268>

02/06/2023 – 06/06/2023 2023 23rd International Conference on Transparent Optical Networks (ICTON), Bucharest, Romania

Reinforcement Learning for Autonomous Traffic Flow Capacity Management

As the dynamicity of the traffic increases, the need for self-network operation becomes more evident. One of the solutions that might bring cost savings to network operators, is that of the dynamic capacity management of large packet flows, especially in the context of packet over optical networks. Machine Learning, and particularly Reinforcement Learning (RL), seem to be an enabler for autonomicity, as a result of its inherent capacity to learn from experience. In this tutorial, we introduce RL and review its application for autonomous capacity management of traffic flows.ion...

Link <https://ieeexplore.ieee.org/document/10207531>

02/06/2023 – 06/06/2023 2023 23rd International Conference on Transparent Optical Networks (ICTON), Bucharest, Romania

Intent-Based Networking for Zero-Touch Optical Networking

The Intent-Based Networking (IBN) paradigm targets at defining high-level abstractions, so network operators can define what are their desired outcomes without specifying how they would be achieved. The latter can be achieved by leveraging network programmability, monitoring and data analytics, as well as the key assurance component. In this paper, we introduce the IBN paradigm and its application to optical networking, highlighting the benefits that Machine Learning (ML) algorithms can provide to IBN. Illustrative examples of intent-based operation are presented for proactive self-configuration and cooperative intent operation.

Link <https://ieeexplore.ieee.org/document/10207264>

02/06/2023 – 06/06/2023 2023 23rd International Conference on Transparent Optical Networks (ICTON), Bucharest, Romania

Using a SNR Digital Twin for Failure Management

The development of Digital Twins to represent the optical transport network might enable multiple applications for network operation, including automation and fault management. In this work, we use GNPpy as a Signal to Noise Ratio (SNR) digital twin of the optical network for failure management applications. The methodology proposed for failure management consists in comparing the QoT measured in the transponders with the one estimated using the digital twin and try to explain detected deviations as changes in the value of input parameters of the Quality of Transmission (QoT) model representing the optical devices, like noise figure (NF) in optical amplifiers (OA) and reduced Optical SNR in the Wavelength Selective Switches (WSS). By applying reverse engineering, the value of those modeling parameters can be estimated as a function of the observed QoT of the lightpaths. Experiments reveal high accuracy estimation of modeling parameters, and results obtained by simulation show large anticipation of soft-failure detection and localization, as well as accurate identification of degradations before they have a major impact on the network.

Link <https://ieeexplore.ieee.org/document/10207330>

05/03/2023 – 09/03/2023 2023 Optical Fiber Communications Conference and Exhibition (OFC), San Diego, CA, USA

Is Intelligence the Answer to Deal with the 5 V's of Telemetry Data?

Telemetry data and big data share volume, velocity, variety, veracity and value characteristics. We propose a distributed telemetry architecture and show how intelligence can help dealing with the 5 V's of optical networks telemetry data.

Link <https://ieeexplore.ieee.org/document/10116324>

03/07/2022 – 06/07/2022 2022 International Conference on Photonics in Switching and Computing (PSC), Toyama, Japan

Distributed and Autonomous Flow Routing Based on Deep Reinforcement Learning

A DRL approach with a specific reward function is proposed for autonomous flow routing operating on multilayer scenarios. Illustrative results reveal that the DRL achieves optimal flow routing in terms of cost and service quality.

Link <https://ieeexplore.ieee.org/document/9850200>

2015

A New Linearly Tunable RF MEMS Varactor with Latching Mechanism for Low Voltage and Low Power Reconfigurable Networks

A new electro-thermally driven high Q tunable RF MEMS capacitor is proposed in this paper. The tuning mechanism of the varactor is chosen to obtain tuning; hence the capacitor has linear tuning range. In order to satisfy the low voltage operation of the varactor electro-thermal actuators were chosen as the drive and latching mechanism. However, those actuators consume lots of power, accordingly, in order to suppress that shortcoming a true mechanically latching mechanism is proposed which considerably decreases the total power consumption of the varactor. In order to investigate important operational aspects of the proposed varactor, different FEM simulations are carried out. The results are as follow; the Q-factor is 43.39 at 1 GHz for a 0.46 pF capacitance value. The tuning ratio is 1.17:1 and self-resonant frequency is 16.61 GHz. The propose varactor works by applying an actuation voltage of 3.1 V and consumes a power of 180 mW for about 2.7 ms for each tuning steps. The proposed varactor of this paper can be a promising choice for low loss and low power RF tunable networks. © 2015, Springer Science+Business Media New York.

2021

Soft Failure Detection, Localization, Identification, and Severity Prediction by Estimating QoT Model Input Parameters

The performance of optical devices can degrade because of aging and external causes like, for example, temperature variations. Such degradation might start with a low impact on the Quality of Transmission (QoT) of the supported lightpaths (soft-failure). However, it can degenerate into a hard-failure if the device itself is not repaired or replaced, or if an external cause responsible for the degradation is not properly addressed. In this work, we propose comparing the QoT measured in the transponders with the one estimated using a QoT tool. Those deviations can be explained by changes in the value of input parameters of the QoT model representing the optical devices, like noise figure in optical amplifiers and reduced Optical Signal to Noise Ratio in the Wavelength Selective Switches. By applying reverse engineering, the value of those modeling parameters can be estimated as a function of the observed QoT of the lightpaths. Experiments reveal high accuracy estimation of modeling parameters, and results obtained by simulation show large anticipation of soft-failure detection and localization, as well as accurate identification of degradations before they have a major impact on the network.

2021

Packet Flow Capacity Autonomous Operation Based on Reinforcement Learning

As the dynamicity of the traffic increases, the need for self-network operation becomes more evident. One of the solutions that might bring cost savings to network operators is the dynamic capacity management of large packet flows, especially in the context of packet over optical networks. Machine Learning, particularly Reinforcement Learning, seems to be an enabler for autonomicity as a result of its inherent capacity to learn from experience. However, precisely because of that, RL methods might not be able to provide the required performance (e.g., delay, packet loss, and capacity overprovisioning) when managing the capacity of packet flows, until they learn the optimal policy. In view of that, we propose a management lifecycle with three phases: (i) a self-tuned threshold-based approach operating just after the packet flow is set up and until enough data on the traffic characteristics are available; (ii) an RL operation based on models pre-trained with a generic traffic profile; and (iii) an RL operation with models trained for real traffic. Exhaustive simulation results confirm the poor performance of RL algorithms until the optimal policy is learnt and when traffic characteristics change over time, which prevents deploying such methods in operators' networks. In contrast, the proposed lifecycle outperforms benchmarking approaches, achieving noticeable performance from the beginning of operation while showing robustness against traffic changes.

2021

Autonomous and Energy Efficient Lightpath Operation Based on Digital Subcarrier Multiplexing

The massive deployment of 5G and beyond will require high capacity and low latency connectivity services, so network operators will have either to overprovision capacity in their transport networks or to upgrade the optical network controllers to make decisions nearly in real time; both solutions entail high capital and operational expenditures. A different approach could be to move the decision making toward the nodes and subsystems, so they can adapt dynamically the capacity to the actual needs and thus reduce operational costs in terms of energy consumption. To achieve this, several technological challenges need to be addressed. In this paper, we focus on the autonomous operation of Digital Subcarrier Multiplexing (DSCM) systems, which enable the transmission of multiple and independent subcarriers (SC). Herein, we present several solutions enabling the autonomous DSCM operation, including: i) SC quality of transmission estimation; ii) autonomous SC operation at the transmitter side and blind SC configuration recognition at the receiver side; and iii) intent-based capacity management implemented through

2022

[Intent-based networking and its application to optical networks \[Invited Tutorial\]](#)

The intent-based networking (IBN) paradigm targets defining high-level abstractions so network operators can define what their desired outcomes are without specifying how they would be achieved. The latter can be achieved by leveraging network programmability, monitoring, and data analytics, as well as the key assurance component. In this tutorial, we introduce the IBN paradigm and its application to optical networking, highlighting the benefits that machine learning (ML) algorithms can provide to IBN. Because the deployment of ML applications requires a specific orchestrator to create ML functions that are connected as ML pipelines, we show an implementation of such an orchestrator. Some challenges and solutions are presented for the generation of accurate synthetic data, proactive self-configuration, and cooperative intent operation. Illustrative examples of intent-based operation and numerical results are presented, and the obtained performance is discussed. © 2009-2012 Optica Publishing Group.

Provisioning of Time-Sensitive and Non-Time-Sensitive Flows With Assured Performance

Near-real-time 6G service operation enabled by distributed intelligence and in-band telemetry

Scenarios for Optical Encryption Using Quantum Keys

Augmented Reality App with AI-based Pervasive Latency Monitoring of RAN and Programmable Metro Packet-Optical Networks

Autonomous Flow Routing Based on Deep Reinforcement Learning

Deployment of Genuine Multi-Agent Pipelines for Near-Real-Time Control of 6G Network Services

Deployment of Multi-Agent System Pipelines for Near-Real-Time Operation of 6G Network Services

Distributed Network Control for QoS Assurance in MultiDomain Networks

Near Real-Time Autonomous Multi-Flow Routing with Dynamic Optical Bypass Management

Centralized and distributed approaches to control optical point-to-multipoint systems near-real-time

Autonomous Flow Routing for Near Real-Time Quality of Service Assurance

Coordination of Radio Access and Optical Transport

Degradation Detection and Severity Estimation by Exploiting an Optical Time and Frequency Digital Twin

Is Intelligence the Answer to Deal with the 5 V's of Telemetry Data?

Intent-based networking and its application to optical networks [Invited Tutorial]

● PROJECTS

01/01/2023 – 31/01/2025

HORIZON-SNS Self-mAnaged Sustainable high-capacity Optical Networks (SEASON), HORIZON-JU-SNS-2022 (G.A. 101096120)

I have been involved in this project.

Link <https://www.season-project.eu/>

01/01/2023 – 31/01/2025

HORIZON-SNS Deep Programmability and Secure Distributed Intelligence for Real-Time End-to-End 6G Networks (DESIRE6G), HORIZON-JU-SNS-2022 (G.A. 101096466)

I have been involved in this project.

Link <https://desire6g.eu/>

30/11/2019 – 30/11/2021

AI-Powered Intent-Based Packet and Optical Transport Networks and Edge and Cloud Computing for Beyond 5G, Ref: PID2020-114135RB-I00, 2021-2024

I have been involved in this national project.

30/11/2018 – 30/11/2021

Beyond 5G - Optical nEtwork continuum, H2020-ICT-2020-2 (G.A. 101016663)

I have been involved in this project.

30/11/2017 – 30/11/2020

METRO High bandwidth, 5G Application-aware optical network, with edge storage, compute and low Latency, H2020-ICT-2016-2 (G.A. 761727)

I have been involved in this project.

30/11/2017 – 30/11/2019

cogniTive 5G application-aware optical metro netWorks Integrating moNitoring, data analyticS and optimization, Ref: TEC2017-90097-R, 2018-2020

I have been involved in this national project.

01/02/2025 – 31/03/2026

CASTIEL 2

I led Training, Twining and Mentoring work package of this project.

01/04/2026 – CURRENT

CASTIEL 3

Currently I am leading the Training coordination and collaboration across NCCs work package in this international project.

Link <https://hpc-portal.eu/>

● HONOURS AND AWARDS

30/11/2010

Full Scholarship – Sahand University of Technology

Full Scholarship to Sahand University of Technology for duration of M.Sc. degree.

The scholarships are awarded based on the national university entrance exam, which separates the students based on their score and determines the university which the student can attend. The students are divided into many university categories such as: State, Azad, Private, Payamnur, online and etc. Part of the students who are eligible to attend state universities are awarded a full scholarship.

30/11/2009

Ranked 12th – Azad university of Urmia

Ranked 12th in overall GPA among 130 students in Azad university of Urmia

30/11/2005

Ranked 3400th

Ranked 3400th (1% top place) among more than 400,000 participants in the national universities' entrance exam for B.Sc.

30/11/2010

Ranked 924th

Ranked 924th (1% top place) among 200,000 participants in the national universities' entrance for M.Sc. Degree in the field of Electrical Engineering.

Awarded 70% of tuition – Azad University of Urmia

Awarded 70% of tuition for four semesters of B.Sc. based on my GPA.

30/11/2023

Best UPC Doctoral Thesis award – Universitat Politècnica de Catalunya

● **ORGANISATIONAL SKILLS**

First Workshop 'Women in telecommunication and optical networks: Highlighting diverse voices in optical networks' (WeInTel), 23th International Conference on Transparent Optical Networks ICTON 2023

I co-organized the first workshop regarding women in telecommunication and optical networks in 23th International Conference on Transparent Optical Networks ICTON 2023.

Link <https://www.gov.pl/web/instytut-lacznosci/icton-2023>

2nd Workshop 'Women in telecommunication and optical networks: Highlighting diverse voices in optical networks' (WeInTel), 24th International Conference on Transparent Optical Networks ICTON 2024

I co-organized the second workshop regarding women in telecommunication and optical networks in 24th International Conference on Transparent Optical Networks ICTON 2024.

Link <https://www.gov.pl/web/instytut-lacznosci/icton-2024>

3rd Workshop 'Women in telecommunication and optical networks: Highlighting diverse voices in optical networks' (WeInTel), 25th International Conference on Transparent Optical Networks ICTON 2025

I co-organized the third workshop regarding women in telecommunication and optical networks in 25th International Conference on Transparent Optical Networks ICTON 2025.

12/07/2026 – 17/07/2026

4th Workshop 'Women in telecommunication and optical networks: Highlighting diverse voices in optical networks' (WeInTel), 26th International Conference on Transparent Optical Networks ICTON 2026

I co-organize the fourth workshop regarding the women in telecommunication and optical networks in 26th International Conference on Transparent Optical Networks ICTON 2026.

Link <https://icton2026.ufe.cz/en/about/>